

```

C:\Windows\system32\cmd.exe
Could Not Find E:\BACKUP\log\doc.log
Sync 2.1 Copyright 2007 Zach Scrivena 2007-12-09
DIRECTORY SYNCHRONIZATION
Log file: "E:\BACKUP\log\doc.log"
Source directory: "D:\doc\"
Target directory: "E:\BACKUP\doc\"
File-matching attributes: {name,size,time,crc}
Source file/directory filter: NOT glob("*.svn")
Target file/directory filter: NOT glob("*.svn")
Filter mode: filename
SUBDIRECTORY: ".\"
No. of source files matched: 0 of 1
No. of unmatched source files to be copied: 1
[C:\ "CSI Journal Vol 2 No.4 2015.pdf" (4.4 Mb)]
SYNCHRONIZATION REPORT
No. of source subdirectories scanned : 1
No. of source files scanned : 1
No. of target files scanned : 0
No. of source files matched (N) : 0
No. of unmatched source files (C) : 1 (1 copied)
No. of unmatched target files/directories (D) : 0 (0 deleted)
Sync is done!
Backup for doc done.
Press any key to continue . . .

```

Figure 3: Updating the backup. A file has been updated

[“Target”] is modified and, by default, the file name, size, last-modified time and a CRC-32 checksum of the file are used to match files. If [“Source”] is a directory, the source and target directories are matched recursively. Matched target files are time-synced and renamed if necessary; unmatched source files are copied to the target directory and unmatched target files and/or directories are deleted. If [“Source”] is a file, source and target files are matched, while ignoring the file name. If the files match, the target file is time-synced and renamed if necessary; if the target does not exist, the source file is copied to the target.

A few switches

Some of the switches that can be used with Sync are:

-s, --simulate	Simulate only; do not modify target
--ignorewarnings	Ignore warnings; do not pause
-l, --log:<“x”>	Create log file x; if x is not specified, sync.yyyyMMdd-HH:mm:ss.log is used
-r, --norecuse	Do not recurse into sub-directories
-n, --noname	Do not use file name for file-matching
-t, --notime	Do not use last-modified time for file-matching
-c, --nocrc	Do not use CRC-32 checksum for file-matching
--time:[x]	Use an x-millisecond time-tolerance for file-matching (0-millisecond time-tolerance is used by default; use --time:1000 or more to avoid mismatches across different file systems)
--rename:[y/n]	Always[y]/never[n] rename matched target files

--synctime:[y/n]	Synchronise the time of matched target files
--overwrite:[y/n]	Overwrite existing target files/directories
--delete:[y/n]	Delete unmatched target files/directories
--force	Equivalent to the combination:--rename:y --synctime:y --overwrite:y --delete:y

Batch script: sync-generic.bat

Given below is a ‘generic’ script that can be invoked by other backup scripts. The reason for using such a script is that the actual backup script ends up with only minimal details and is thus easier to maintain. Additionally, changes to the commands, if any, when done in a central script work for all dependent scripts, reducing maintenance efforts.

```

@echo off
set SYNC_HOME=.\Sync.jar
rem deleting existing log file
del log\%1.log
java -jar Sync.jar --log:log\%1.log --force %2 %3
echo Backup for %1 done.

```

Batch script: backup-doc.bat

Given below is the script that takes a backup of the ‘doc’ directory and uses the generic script for this purpose.

```

@echo off
set NAME=doc
call sync-generic.bat %NAME% D:\%NAME% .\%NAME%
pause

```

While there are many personal backup solutions, my tool of choice is Sync – a Java based command-line tool. Though it does not have a fancy GUI, it helps me maintain backups with little effort. To make the task of backing up easier, I use Windows batch files, which only require me to double-click (or press an *Enter* key) the script, to kick off the backup process. 

References

- [1] Syncdir, <http://syncdir.sourceforge.com/>
- [2] Best free backup software: <http://www.in.techradar.com/news/software/applications/Best-free-backup-software-11-programs-we-recommend/articleshow/38877922.cms>
- [3] 13 Best backup software: <http://www.pcadvisor.co.uk/test-centre/software/13-best-backup-software-2015-2016-uk-3263573/>

By: Bipin Patwardhan

The author is a senior technical architect at Capgemini India. He believes in the value of open source and has developed many customer solutions using open source Java libraries like Apache XMLBeans.